

Report on Protein Secondary Structure Prediction

Project: Protein Secondary Structure Prediction (Q8 & Q3)

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Dataset Preparation

The dataset consists of protein sequences in Q8 and Q3 formats with corresponding secondary structure labels.

- Each protein sequence is represented by a string of amino acids.
- Q8 designations correspond to eight secondary structural classes.
- Q3 labels are produced by combining Q8 classifications into three more broad categories (Helix, Sheet, and Coil).
- A special PyTorch Dataset class was developed to:
 - Encrypt amino acids using a vocabulary mapping (stoi).
 - Be cautious when handling unknown residues.
 - Sequences are shortened to a maximum length that is preset.

Preprocessing

To make variable-length sequences suitable for batch-wise training:

- Integer indices were created from amino acids.
- A unique PAD token was used to apply padding.
- A unique collate_fn was created in order to:
 - Within each batch, dynamically pad sequences.
 - Use ignore_index = -100 to mask padded positions when calculating the loss.
 - To guarantee that only valid residues were evaluated, sequence lengths were maintained.

Model Architecture (Scratch Models)

Models that are trained in notebook:

- Bi-RNN
- Bi-GRU
- CNN-LSTM (for better accuracy)

Training Configuration

- CrossEntropyLoss using:
 - To conceal padded positions, set ignore_index = -100.
 - For improved generalization, use label smoothing.
- Handling Class Imbalance: Frequency-based inverse weighting was used to calculate the Q8 class weights.
- Optimizer: Adam optimizer
- Learning Rates: For smooth decay, use CosineAnnealingLR.
- Clipping Gradients: stopped gradients in recurrent layers from exploding

Variants of the Model Tested

Several architectural variations were investigated in order to enhance performance:

- deepening the GRU (two to three layers).
- After recurrent layers, add residual connections.
- Loss weighting is adjusted to give Q8 performance priority.
- experimenting with various dropout rates & hidden layers.
- Attention-based pooling was tried, but it was later eliminated because it didn't match token-level prediction.
- At last, added CNN-LSTM to improve accuracy

Results

Model Variant	F1 (Q8)	F1 (Q3)	HM
BiRNN	~ 0.30	~ 0.689	~ 0.43
BiGRU	~ 0.32	~ 0.69	~ 0.44
With Class Weighting (Q8) & Adding a hidden layer	~ 0.33 – 0.3350	~ 0.700 - 0.710	~ 0.456
CNN – LSTM	~ 0.34 – 0.35	~ 0.71 – 0.715	~ 0.46